

Assignment 5

①

$$\sigma_{y+} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad \sigma_{y-} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\sigma_{x+} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \sigma_{x-} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$U = \langle \text{new} \times | \text{old} \rangle$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} (1 \ 1) \begin{pmatrix} 1 \\ i \end{pmatrix} & \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} (1 \ 1) \begin{pmatrix} 1 \\ -i \end{pmatrix} \\ \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} (1 \ -1) \begin{pmatrix} 1 \\ i \end{pmatrix} & \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} (1 \ -1) \begin{pmatrix} 1 \\ -i \end{pmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} (1+i) & \frac{1}{2} (1-i) \\ \frac{1}{2} (1-i) & \frac{1}{2} (1+i) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$

② Complete basis is defined by outer product of each vectors sum to identity matrix.

$$\begin{aligned} \sum_{i=0}^{n-1} |i\rangle\langle i| &= |-\rangle\langle -| + |+\rangle\langle +| \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \frac{1}{\sqrt{2}} (1 \ -1) + \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} (1 \ 1) \\ &= \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

③ $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$\begin{aligned} \sigma_x &= \sum_{i=0}^{2-1} \lambda_i |i\rangle\langle i| \\ &= \underbrace{1 \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \left\langle \frac{1}{\sqrt{2}} (1 \ 1) \right\rangle \right)}_{|+\rangle\langle +|} + \underbrace{-1 \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \left\langle \frac{1}{\sqrt{2}} (1 \ -1) \right\rangle \right)}_{-|-\rangle\langle -|} \\ &= \frac{1}{2} \begin{pmatrix} 1-1 & 1+1 \\ 1+1 & 1-1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

⑤ $|f\rangle_1 \otimes |g\rangle_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \\ 1 \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$$|g\rangle_1 \otimes |f\rangle_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Both are not same.

⑥ $\underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}}_{4 \times 1} \otimes \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{2 \times 1}$

$$= \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}_{4 \times 1} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}_{2 \times 1}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}_{8 \times 1}$$

$$\begin{aligned}
 \textcircled{9}. (|f\rangle + |e\rangle) \otimes |g\rangle &= |f\rangle \otimes |g\rangle + |e\rangle \otimes |g\rangle \\
 &= \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} i \\ 0 \end{pmatrix} \right) \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} + \begin{pmatrix} i \\ 0 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \\
 &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} + \begin{pmatrix} \frac{1}{\sqrt{2}} \\ i/\sqrt{2} \\ 0 \\ 0 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}
 \end{aligned}$$

$$\textcircled{9}. \langle h_1 | h_2 \rangle = \langle \sum_{j=1}^{n_1} |f_j\rangle \otimes |g_j\rangle | \sum_{k=1}^{n_2} |e_k\rangle \otimes |k_1\rangle = \sum_{j=1}^{n_1} \sum_{k=1}^{n_2} \langle f_j | e_k \rangle \langle g_j | k_1 \rangle$$

$$\text{L.H.S} = \langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} i \\ 1 \end{pmatrix} | \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} i \\ 1 \end{pmatrix} \rangle$$

$$= \underline{(-i - i - i - i)} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} = 1 - i$$

$$\text{R.H.S} = (1 \ 1) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cdot (-i \ -i) \begin{pmatrix} i \\ 1 \end{pmatrix}$$

$$= (0 + 1) \cdot (1 - i)$$

$$= 1 - i$$

$$\begin{aligned}
 10 \quad \sigma_x |f\rangle_1 \otimes \sigma_x |g\rangle_2 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \\
 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} i/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \\
 &= \begin{pmatrix} i/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{pmatrix}
 \end{aligned}$$

$$|f\rangle_1 \otimes |g\rangle_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\sigma_x \otimes \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$(\sigma_x \otimes \sigma_x) (|f\rangle \otimes |g\rangle_2) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{pmatrix}$$

equal